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TFM Validation and Experiment Protocol

Scenario Matrix, Metrics, and Trace Evidence

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Repository:	https://github.com/ieverythng/nao-ros4hri-bridge

TFM Validation and Experiment Protocol

Purpose

This document converts the implementation into a thesis validation protocol. It is intended to support the methodology, results, and discussion chapters by making experiments repeatable and by naming the evidence that each run must collect.

Validation Thesis

The system should be evaluated not only by whether the robot completes a task, but by whether the architecture preserves the intended boundaries while doing so. A successful planner-mediated run should show correct routing, valid contract generation, deterministic admission, skill dispatch, feedback publication, and a single user-facing speech authority.

Research Questions Covered

Research question	Observable evidence
Can planner-mediated execution improve multi-step task handling?	Plan validity, step completion, replanning events
Does the skill registry reduce invalid robot-specific outputs?	Invalid-step rate, unsupported-skill rejection rate
Does execution feedback improve recovery?	Replan success, clarification success, failure explanation quality
Do architecture boundaries remain inspectable?	Trace completeness, contract validity, ownership violations
What limitations remain under real HRI constraints?	Latency, duplicate speech, grounding staleness, model failure cases

Scenario Matrix

Scenario group	Example command	Expected behavior	Primary metrics
Dialogue-only	hello	Remains dialogue, no planner request	false planner-route rate
Knowledge query	what can you see?	Uses KB/scene answer, no unnecessary execution	route accuracy, answer groundedness
Atomic execution	look left	Single validated skill step	completion, latency
Grounded look-at	look at the person	Uses visible person target and `look_at` step	target correctness, person/object separation
Multi-step task	scan for a cup and tell me what you found	Planner emits ordered steps	plan validity, step success
Failure recovery	target disappears before execution	Feedback triggers replan or failure dialogue	recovery rate
Clarification	ambiguous target	Planner asks for missing slot	clarification precision
Cancellation/update	stop or changed goal	Goal transition is explicit	supersede/cancel correctness

Runtime Evidence to Capture

- user utterance and timestamp;

- chatbot route and user_intent JSON;
- planner request envelope and Intent.data;
- grounded context snapshot (knowledge_snapshot, scene_summary, state_t0);
- planner output JSON;
- orchestrator validation result;
- dispatched action server and step id;
- execution feedback JSON;
- planner dialogue act, if emitted;
- final user-facing utterance and speech owner.

Quantitative Metrics

Metric	Definition	Interpretation
Route accuracy	Fraction of turns routed to the intended class	Tests chatbot planner gating
Plan validity	Fraction of planner outputs accepted by orchestrator	Tests planner contract discipline
Unsupported step rate	Fraction of steps rejected as unknown or invalid	Tests registry grounding
Task completion	Fraction of scenarios completed	Tests end-to-end behavior
Recovery success	Fraction of failures resolved by replan or clarification	Tests closed-loop supervision
Duplicate speech rate	Fraction of turns with multiple competing utterances	Tests dialogue ownership
Grounding correctness	Fraction of selected targets matching scene/KB evidence	Tests perception-to-planner context
Median latency	Time from user utterance to first action or answer	Tests HRI usability

Qualitative Analysis Dimensions

Each trace should explain why the chatbot selected the route, what grounded context was available at T0, why the planner selected the steps, whether the orchestrator accepted or rejected each step, how feedback changed planner state, whether the final user-facing response had one owner, and what the trace reveals about limitations of the architecture.

Ablation Plan

Ablation	Change	Hypothesis
Direct execution only	Disable planner-mediated path	Multi-step and failure recovery become less robust
No grounded context	Remove `knowledge_snapshot`, `scene_summary`, `state_t0`	Target selection and perception tasks degrade
No execution feedback	Suppress feedback to planner	Replanning and failure explanations degrade

Ablation	Change	Hypothesis
Minimal registry	Hide some skill metadata	Unsupported or underspecified plan steps increase
Chatbot wording disabled	Direct planner dialogue only	Completion wording may become less conversational but easier to attribute

Validation Gates Before Experiments

- Run contract unit tests for planner_common, chatbot_llm, planner_llm, and nao_orchestrator.
- Run registry consistency checks so planner-visible skills match canonical AB entries.
- Run ROS4HRI structural audit to confirm changed packages and ownership-sensitive files.
- Launch the sim profile and verify expected action servers are available.
- Execute one dialogue-only, one knowledge-query, one atomic action, and one failure case before collecting formal results.

Experiment Log Template

Field	Value
Experiment ID	`EXP-YYYYMMDD-###`
Scenario group	
User utterance	
Expected route	
Observed route	
Planner request published	yes/no
Planner output valid	yes/no
Executed steps	
Feedback events	
Final speech owner	
Outcome	success / failure / partial
Notes	

Threats to Validity

The main internal threats are prompt sensitivity, incomplete test coverage for live ROS timing, and possible mismatch between simulated and robot execution. External threats include model availability, cloud/local model differences, and the fact that the validation environment is centered on the NAO stack. Construct validity depends on whether metrics such as plan validity and route accuracy genuinely capture user-perceived task success. The thesis should therefore combine quantitative tables with trace-based qualitative analysis.

Expected Results Narrative

A strong thesis result would show that planner-mediated execution does not merely add complexity. It should demonstrate that the planner path improves inspectability, makes multi-step execution more explicit, and creates recovery opportunities through feedback. Even when the planner fails, the architecture should make the failure attributable: wrong route, stale grounding, invalid plan, missing skill, execution failure, or dialogue ownership issue.